Back

Eigenvalues and Eigenvectors Graded Assignment
Graded Assignment • 10 min

Due Nov 9, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. **True** or **False**: An $n \times n$ matrix A can have no more than n eigenvalues. 1 / 1 point

   - (•) True
   - ( ) False

   Nice work

2. **True** or **False**: If $\lambda \neq 0$ is an eigenvalue of an invertible $n \times n$ matrix A, then λ^{-1} is an eigenvalue of the matrix A^{-1}. 1 / 1 point

   - (•) True
   - ( ) False

   Nice work

3. **True** or **False**: If $\mathbf{u}$ and $\mathbf{v}$ are both eigenvectors of an $n \times n$ matrix A, then $\mathbf{u} + \mathbf{v}$ is also an eigenvector of A. 1 / 1 point

   - ( ) True
   - (•) False

   Nice work

4. **True** or **False**: Suppose the $n \times n$ matrix A has n distinct eigenvalues. Then the dimension of each eigenspace is 1. 1 / 1 point

   - (•) True
   - ( ) False

   Nice work

5. **True** or **False**: If $p(\lambda) = \det(A - \lambda I_n)$ is the characteristic polynomial of an $n \times n$ matrix A, and $p(0) = 0$, then A is not invertible. 1 / 1 point

   - (•) True
   - ( ) False

   Nice work